

**University of Toronto Scarborough Campus**  
**Department of Computer and Mathematical Sciences**

These solutions are not the only “perfect” or “10/10” solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

## Problems

Q1. Determine if  $\mathbf{v}_1$  can be written as a linear combination of  $\mathbf{v}_2$  and  $\mathbf{v}_3$  for the following  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ . If it is possible to write  $\mathbf{v}_1$  as a linear combination of  $\mathbf{v}_2$  and  $\mathbf{v}_3$ , give the linear combination. If it is impossible, justify why it is impossible.

(a)

$$\mathbf{v}_1 = (-2, 0, 3) \quad \mathbf{v}_2 = (1, 3, 0) \quad \mathbf{v}_3 = (2, 4, -1)$$

**Solution:** Let's assume that  $a$  and  $b$  are real numbers such that

$$\begin{pmatrix} -2 \\ 0 \\ 3 \end{pmatrix} = a \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + b \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}$$

This induces the system of linear equations

$$\begin{array}{rcl} a & +2b & = -2 \\ 3a & +4b & = 0 \\ & -b & = 3 \end{array}$$

By the third equation,  $b = -3$ , and by replacing the value of  $b$  in the first and second equations, we obtain that  $a = 4$ , so we can conclude that

$$\begin{pmatrix} -2 \\ 0 \\ 3 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}$$

(b)

$$\mathbf{v}_1 = x^3 - 3x + 5 \quad \mathbf{v}_2 = x^3 + 2x^2 - x + 1 \quad \mathbf{v}_3 = 3x^3 - 6x^2 + x + 4$$

**Solution:** Once again, to find real numbers (if they exist)  $a$  and  $b$  such that  $\mathbf{v}_1 = a\mathbf{v}_2 + b\mathbf{v}_3$ , we need to understand the possible solutions to the system of linear equations

$$\begin{array}{rcl} a & +3b & = 1 \\ 2a & -6b & = 0 \\ -a & +b & = -3 \\ a & +4b & = 5 \end{array}$$

Nevertheless, note that by adding the third and fourth equations, we obtain that  $b = 2/5$ , and by subtracting 2 times the first equation from the second one, we obtain that  $b = 1/12$ , which is a contradiction.

(c)

$$\mathbf{v}_1 = \begin{bmatrix} 5 & 1 \\ 6 & -5 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 1 & -12 & -1 \end{bmatrix}$$

**Solution:** Note that  $\mathbf{v}_2$  and  $\mathbf{v}_3$  are not even elements of the same vector space.

Q2. Find a linear system whose solution set is given by:

$$W = \text{span} \{(1, 1, 2, 3), (1, 4, 2, 1)\}$$

**Solution:** Since every solution to the system of linear equations that we're trying to find is in  $W$ , we have that a vector  $(x_1, x_2, x_3, x_4)$  is a solution to this system if, and only if, there are real numbers  $s$  and  $t$  such that

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = s \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 4 \\ 2 \\ 1 \end{pmatrix}$$

This induces the system of linear equations

$$\begin{aligned} x_1 &= s + t \\ x_2 &= s + 4t \\ x_3 &= 2s + 2t \\ x_4 &= 3s + t \end{aligned}$$

From here, we can deduce, for example, that

$$x_3 - 2x_1 = 0.$$

Also,

$$x_2 - x_1 = 3t \quad \text{and} \quad 2x_4 - 3x_3 = -4t,$$

which implies that

$$4(x_2 - x_1) + 3(2x_4 - 3x_3) = 0.$$

Since the system of linear equations that we found

$$\begin{array}{rrrr} -2x_1 & & +x_3 & = 0 \\ -4x_1 & +4x_2 & -9x_3 & +6x_4 = 0 \end{array}$$

has two free variables, its set of solutions is  $W$ .

Q3. Solve the following linear systems.

$$(a) \begin{cases} x_1 - 3x_2 = 1 \\ 2x_1 + 4x_2 = 2 \end{cases}$$

**Solution:** By subtracting two times the first equation from the second one, we obtain that  $x_2 = 0$ , and therefore  $x_1 = 1$ .

$$(b) \begin{cases} x_1 + 4x_2 = 0 \\ 2x_1 + 3x_2 = 1 \\ -x_1 + 6x_2 = -2 \end{cases}$$

**Solution:** By adding the first and third equations, we have that  $x_2 = -1/5$ . By replacing the value of  $x_2$  in all of the three equations, we can conclude that  $x_1 = 4/5$ .

Q4. Find a set of vectors that spans the subspace  $W \subseteq V$  of solutions to the following linear systems.

$$(a) \begin{cases} x_1 + 2x_2 - x_3 + x_4 = 0 \\ -3x_1 + x_3 + 2x_4 = 0 \end{cases}$$

**Solution:** Let's consider the extended matrix associated to the system

$$\left( \begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ -3 & 0 & 1 & 2 & 0 \end{array} \right)$$

By applying the elementary operation  $R_2 + 3R_1$  we obtain the matrix

$$\left( \begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 6 & -2 & 5 & 0 \end{array} \right)$$

Applying now the elementary operation  $3R_1 - R_2$  we obtain

$$\left( \begin{array}{cccc|c} 3 & 0 & -5 & -2 & 0 \\ 0 & 6 & -2 & 5 & 0 \end{array} \right)$$

Finally, we can apply the operations  $R_1/3$  and  $R_2/6$  to obtain the echelon form

$$\left( \begin{array}{cccc|c} 1 & 0 & -5/3 & -2/3 & 0 \\ 0 & 1 & -1/3 & 5/6 & 0 \end{array} \right)$$

Now, this extended matrix has the associated system of linear equations

$$\begin{array}{rrcr} x_1 & & -5/3x_3 & -2/3x_4 & = 0 \\ & x_2 & -1/3x_3 & 5/6x_4 & = 0 \end{array}$$

By parametrizing  $x_3 = s$  and  $x_4 = t$ , we can conclude that every solution to the original system of equations is of the form

$$\begin{pmatrix} 5/3s + 2/3t \\ 1/3s - 5/6t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 5/3 \\ 1/3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2/3 \\ 5/6 \\ 0 \\ 1 \end{pmatrix}$$

Therefore, the set of solutions to the system of equations above is the span of the vectors

$$\begin{pmatrix} 5/3 \\ 1/3 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2/3 \\ 5/6 \\ 0 \\ 1 \end{pmatrix}.$$

(b)  $\{x_1 + 2x_3 + x_4 = 0\}$

**Solution:** In this case, we can parametrize the variables  $x_2 = r$ ,  $x_3 = s$ ,  $x_4 = t$ , and conclude that every solution to the system above is of the form

$$\begin{pmatrix} -2s - t \\ r \\ s \\ t \end{pmatrix} = r \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Therefore, the set of solutions for the system above is the span of the vectors

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

In the next two questions, you will explore two important subspaces associated to a system of linear equations. We will revisit these two subspaces later on in the course. Presently, the notation will be quite cumbersome and difficult. When we develop more theory, the following two questions will have very simple proofs.

Q5. Suppose that  $\mathbf{v} = (x_1, x_2, \dots, x_n)$  and  $\mathbf{v}' = (x'_1, x'_2, \dots, x'_n)$  are solutions of the system:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{k1}x_1 + a_{k2}x_2 + \cdots + a_{kn}x_n = 0 \end{cases}$$

Prove that the following are also solutions to the system.

(a)  $\lambda \mathbf{v}$  for  $\lambda \in \mathbb{R}$

**Solution:** For each  $i \leq k$ , note that

$$a_{i1}(\lambda x_1) + a_{i2}(\lambda x_2) + \cdots + a_{in}(\lambda x_n) = \lambda(a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n) = 0$$

given that  $\mathbf{v} = (x_1, \dots, x_n)$  is a solution to the system. Since the equality above is true for every  $i \leq k$ , we can conclude that  $\lambda \mathbf{v}$  is a solution to the system as well.

(b)  $\mathbf{v} + \mathbf{v}'$

**Solution:** Once again, for each  $i \leq k$  we have that

$$\begin{aligned} a_{i1}(x_1 + x'_1) + a_{i2}(x_2 + x'_2) + \cdots + a_{in}(x_n + x'_n) \\ = (a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n) + (a_{i1}x'_1 + a_{i2}x'_2 + \cdots + a_{in}x'_n) = 0 \end{aligned}$$

given that both  $\mathbf{v} = (x_1, \dots, x_n)$  and  $\mathbf{v}' = (x'_1, \dots, x'_n)$  are solutions to the system. Since the equality above is true for all  $i \leq k$ ,  $\mathbf{v} + \mathbf{v}'$  is a solution to the system as well.

Conclude that the set of solutions is a subspace of  $\mathbb{R}^n$ .

Q6. Suppose that the linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = c_{1i} \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = c_{2i} \\ \vdots \\ a_{k1}x_1 + a_{k2}x_2 + \cdots + a_{kn}x_n = c_{ki} \end{cases}$$

has a solution  $\mathbf{v}_i = (b_{i1}, b_{i2}, \dots, b_{in})$  for each  $\mathbf{c}_i = (c_{1i}, c_{2i}, \dots, c_{ki})$  in  $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ . Prove that the linear system has a solution  $\mathbf{v}$  for each  $\mathbf{c}$  in  $\text{span}(\mathbf{c}_1, \dots, \mathbf{c}_n)$ .

**Solution:** Given  $\mathbf{c} \in \text{span}(\mathbf{c}_1, \dots, \mathbf{c}_n)$ , let  $\lambda_1, \dots, \lambda_n$  be real numbers such that

$$\mathbf{c} = \lambda_1 \mathbf{c}_1 + \cdots + \lambda_n \mathbf{c}_n$$

and consider the vector  $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n$ . We claim that the system obtained by putting the entries of the vector  $\mathbf{c}$  on the right hand side of the equations above, has the vector  $\mathbf{v}$  as a solution: for each  $j \leq k$ , if we replace the variables in the system by the entries of the vector  $\mathbf{v}$ , we have that

$$\begin{aligned} & a_{j1}(\lambda_1 b_{11} + \lambda_2 b_{21} + \cdots + \lambda_n b_{n1}) + \cdots + a_{jn}(\lambda_1 b_{1n} + \lambda_2 b_{2n} + \cdots + \lambda_n b_{nn}) \\ &= \lambda_1(a_{j1}b_{11} + \cdots + a_{jn}b_{1n}) + \cdots + \lambda_n(a_{j1}b_{n1} + \cdots + a_{jn}b_{nn}) \\ &= \lambda_1 c_{j1} + \cdots + \lambda_n c_{jn}. \end{aligned}$$

But note that the number  $\lambda_1 c_{j1} + \cdots + \lambda_n c_{jn}$  is exactly the  $j$ th entry of the vector  $\mathbf{c}$  (!) as desired.

In the following question, you will be asked why our textbook defines operations in a particular way. Your answer to this question should be a mix of mathematics and common sense. Explain why the textbook authors made the choices that they did when writing the book.

Q7. In Proposition 1.5.3 (pg. 35) of A Course in Linear Algebra by Damiano and Little, there are three row operations. The operation (b) of scaling requires the scalar to be a non-zero number. Why does it the number to be non-zero? Similarly, operation (a) of adding two rows requires the rows to be distinct. Why are the rows required to be distinct?

**Solution:** Otherwise, we would be able to transform any matrix into the matrix whose entries are all 0 (either by multiplying each row by zero or subtracting a row from itself). This is problematic since one of the main properties of elementary operations is that the set of solutions is preserved by applying them. If we change the set of solutions when applying operations to a system (or an extended matrix) then we are not really solving anything.

One of the super-powers of linear algebra is that it can be applied everywhere. In the next question, you will use linear algebra to solve a non-linear problem.

- Q8. Given a list of  $n$  points  $(x_i, y_i) \in \mathbb{R}^2$  with distinct  $x_i$  values, it is sometimes necessary to find a polynomial curve  $p(x) \in \mathbf{P}_{n-1}(\mathbb{R})$  passing through all the points. One method to do so, uses linear algebra as follows. (We will learn another method, called Lagrange interpolation, later in the course.)

Consider an arbitrary polynomial  $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1}$ . We consider the coefficients as unknown. Each point  $(x_i, y_i)$  gives a linear relation among the coefficients. The linear system is given:

$$\begin{cases} a_0 + a_1x_1 + a_2x_1^2 + \cdots + a_{n-1}x_1^{n-1} &= y_1 \\ a_0 + a_1x_2 + a_2x_2^2 + \cdots + a_{n-1}x_2^{n-1} &= y_2 \\ &\vdots \\ a_0 + a_1x_n + a_2x_n^2 + \cdots + a_{n-1}x_n^{n-1} &= y_n \end{cases}$$

- (a) Use this method to find a quadratic polynomial passing through the points  $(0, 3)$ ,  $(1, 2)$ ,  $(2, 3)$ .

**Solution:** Following the method described above, we're looking for a solution to the system

$$\begin{array}{rrrr} a_0 & & & = 3 \\ a_0 & +a_1 & +a_2 & = 2 \\ a_0 & +2a_1 & +4a_2 & = 3 \end{array}$$

Clearly,  $a_0 = 3$ , and by subtracting 4 times the second equation from the third, we obtain that  $a_1 = -2$ , and therefore  $a_2 = 1$ . Thus, the desired polynomial is  $x^2 - 2x + 3$ .

- (b) Use this method to find a cubic passing through the points  $(0, 5)$ ,  $(1, 9)$ ,  $(2, 11)$ ,  $(3, 5)$ .

**Solution:** Once again, we're looking for a solution to the system

$$\begin{array}{rrrrr} a_0 & & & & = 5 \\ a_0 & +a_1 & +a_2 & +a_3 & = 9 \\ a_0 & +2a_1 & +4a_2 & +8a_3 & = 11 \\ a_0 & +3a_1 & +9a_2 & +27a_3 & = 5 \end{array}$$

which is given by  $a_0 = 5$ ,  $a_1 = 3$ ,  $a_2 = 2$ , and  $a_3 = -1$ . Therefore, the polynomial

$$-x^3 + 2x^2 + 3x + 5$$

is as required.

- (c) Consider the example of the three points laying on a line, such as:  $(0, 0)$ ,  $(1, 1)$ ,  $(2, 2)$ . What happens to your linear system in this case? In general, what happens to this method if the points lie on a curve of a lower degree than necessary?

**Solution:** Following the method described above, we will end up with the system of equations

$$\begin{array}{rrrr} a_0 & & & = 0 \\ a_0 & +a_1 & +a_2 & = 1 \\ a_0 & +2a_1 & +4a_2 & = 2 \end{array}$$

Since  $a_0 = 0$ , we can reduce the system to

$$\begin{array}{rcl} a_1 & +a_2 & = 1 \\ 2a_1 & +4a_2 & = 2 \end{array}$$

which implies that  $a_1 = 1$  and  $a_2 = 0$ , so the polynomial that we obtain is indeed  $x$ .